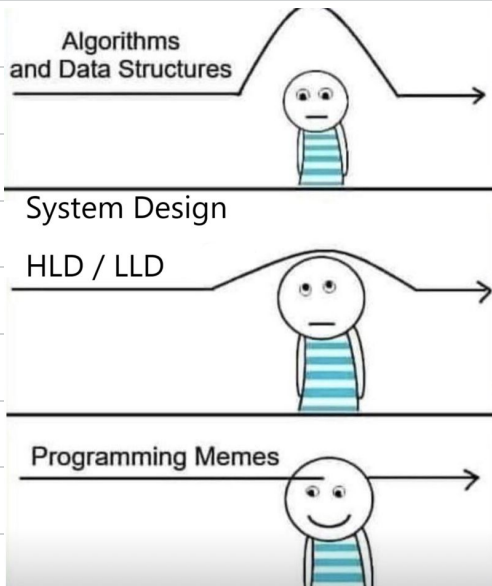




My brain when I am sitting in a system design interview



Agenda

- Overview of Netflix Recommendation system-design



Difference between system design for Machine Learning Engineers and Software Developers.

Aspect	ML Engineer	SOE
Primary goal	Accurate / Fast prediction stable data flow / feature pipeline	Reliable feature delivery, interconnectivity diff of software, Accurate API struct
Artifacts	Models, Feature, training Code;	API, Service, DB
Variability	Drifts	API Req. / Response Validation
Metrics	Offline (Online metr	Latency, Throughput

Where do we start??

Mlops — CLARIFYING.

① Expected # of Users
1 Billion Users
DAU 200 million Users

② Expected no. of Videos
1 million

③ # of Videos to be uploaded : 1000

④ # of times user accesses netflix : 5

"Password Incorrect"

"Password Incorrect"

resets password

"Your password cannot be
your previous password"



Why UX research is important



$$\frac{200 \times 10^6 \times 5}{24 \times 60 \times 60} = 12K \text{ req. per second.}$$

LLD $\rightarrow$ low level design $\rightarrow$ Implementation detail

HL D $\rightarrow$ High level design $\rightarrow$ architecture details

2022 $\rightarrow$ GPT

Accuracy Vs Latency

1 FUNCTIONAL REQUIREMENTS

1. Users should be able to stream video on the platform (pause/resume video playback) .
2. Users should be able to rate/like/dislike/comment on a video/movie.
3. Personalized, and non-personalized recommendation.
4. Users should be able to search video on the platform.
5. Need to provide different video quality options.
6. Producers, directors to be able to upload video on the platform.
7. Should be available on platforms like iOS/Android/Web.

2 NON-FUNCTIONAL REQUIREMENTS

- ① V. low latency & high availability.
- ② Scalable & Efficient
- ③ Slick UI / easy to interact with.

In the context of system design, what refers to the overall view of the system, including architecture, applications, and database management?

0 users have participated

A

Low-Level Design (LLD)

0%

B

High-Level Design (HLD)

0%

C

System Data Flow

0%

D

Component Execution

0%

End Quiz Now

Based on all quizzes from the session

Sanidhya Shukla

1/1 ⚡ 94.45

Rishikesh Kumar ...

1/1 ⚡ 98.90

Yadwinder kaur

1/1 ⚡ 89.77

4	Vidhyashankar M	1/1 ⚡ 85.42
5	J. Ramesh Reddy	1/1 ⚡ 82.83
6	Jigar	1/1 ⚡ 82.60
7	Manikandan A	1/1 ⚡ 79.03

What is an essential aspect of system design in regards to understanding its scale?

6 users have participated

A

Estimating the color palette for the UI

0%

B

Programming language to be used

0%

C

Making reasonable assumptions about system metrics like requests per day or database calls

100%

D

Selecting a font style for documentation

0%

End Quiz Now

Rishikesh Kumar ...

2/2 ⚡ 190.20

Sanidhya Shukla

2/2 ⚡ 193.09

Yadwinder kaur

2/2 ⚡ 181.36

4	Vidhyashankar M	2/2 ⚡ 175.00
5	J. Ramesh Reddy	2/2 ⚡ 171.90
6	Manikandan A	2/2 ⚡ 170.76
7	Jigar	2/2 ⚡ 161.37
8	K. Janani	1/2 ⚡ 94.17
9	shreyas	1/2 ⚡ 91.44
10	Kuntal Kole	1/2 ⚡ 87.60

Different components in our MVP

- ✓ 1. Data Sources.
- ✓ 2. Data Ingestion Pipeline
- ✓ 3. Data Storage Systems.
- ✓ 4. Data Processing and Feature Engineering.
- 5. Recommendation Engine.
- 6. API Layer.
- ✓ 7. Caching
- ✓ 8. Monitoring and Logging.
- 9. Security and Compliance.

① Data Sources

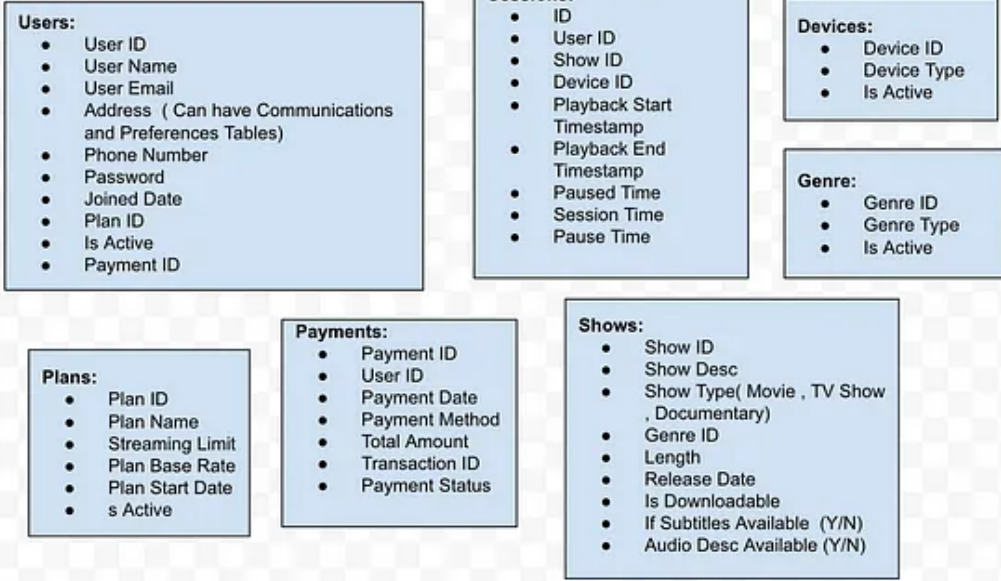
② Users / User-item Interaction.

- (i) Like / dislike / review / ratings / comments
- (ii) Watch history
- (iii) Age, gender, demographics, Language
- (iv) User Interaction
- (v) Wish list / watch list
- (vi) Daily time spent

③ Content DB

- (i) Movies / Shows (Genre, director, casts, title, duration, ratings, Kids, 18+ description,
- (ii) Release Year
- (iii) Technical specification - 4K / 1080p 24p...
- (iv) Content popularity metrics → IMDB, Rotten tomato

(w) External data → ① Social media friends
 ② Famous third party reviews



2

DATA PIPELINE

Client Side

Server Side,

click stream data

Recommendation results

Device Usage data

Which result was click on.

Wakeup/Shut down time

click - 1
add to cart - 2
checkout - 3
proceed to payment - 4
successful payment - 5

Non-structured or structured data base for our POC?

Structured DD

```
// Electronics Product
{
  "product_id": 1,
  "name": "Smartphone X",
  "category": "Electronics",
  "price": 999,
  "attributes": {
    "batteryLife": "24h",
    "warrantyPeriod": "2 years",
    "brand": "TechBrand"
  },
  "reviews": [
    {
      "user_id": 101,
      "rating": 5,
      "comment": "Excellent phone with great battery life!"
    },
    {
      "user_id": 102,
      "rating": 4,
      "comment": "Good performance but a bit pricey."
    }
  ]
}

// Clothing Product
{
  "product_id": 2,
  "name": "T-Shirt Y",
  "category": "Clothing",
  "price": 29,
  "attributes": {
    "size": "L",
    "material": "Cotton",
    "gender": "Unisex"
  },
  "reviews": [
    {
      "user_id": 103,
      "rating": 4,
      "comment": "Comfortable and fits well."
    }
  ]
}

// Books Product
{
  "product_id": 3,
  "name": "Novel Z",
  "category": "Books",
  "price": 15,
  "attributes": {
    "author": "Author A",
    "ISBN": "123-4567890",
    "genre": "Fiction"
  },
  "reviews": [
    {
      "user_id": 104,
      "rating": 5,
      "comment": "A gripping novel that kept me up all night!"
    }
  ]
}
```


What is the main difference between horizontal scaling and vertical scaling?

0 users have participated

- A Horizontal scaling adds more machines, while vertical scaling adds new software. 0%
- B Horizontal scaling adds physical locations, while vertical scaling focuses on security enhancements. 0%
- C Horizontal scaling adds resources to a single machine, while vertical scaling adds more machines. 0%
- D Horizontal scaling adds more machines, while vertical scaling adds resources to a single machine. 0%

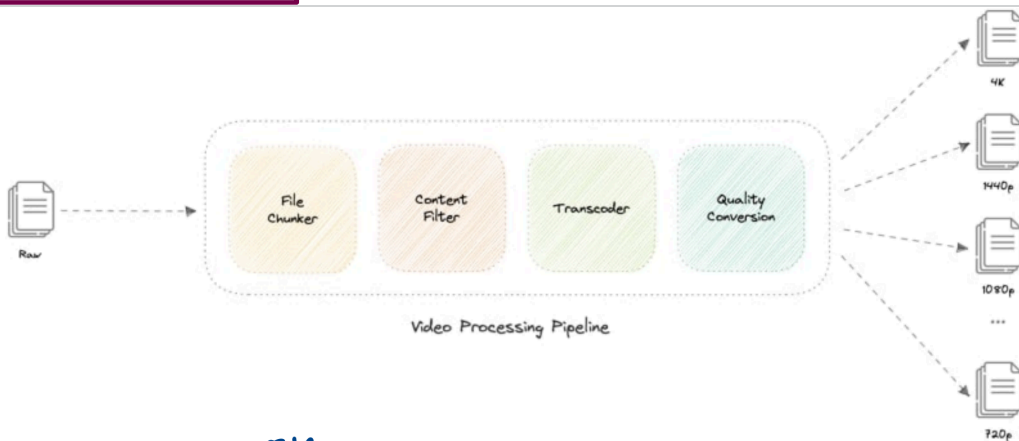
End Quiz Now

Based on all quizzes from the session

R			S			S		
Rishikesh Kumar ...			Sanidhya Shukla			Vidhyashankar M		
3/3	288.06		3/3	288.99		3/3	262.37	
4	Jigar	3/3	231.20					
2	Yashvinder kaur	2/3	181.36					
6	J. Ramesh Reddy	2/3	171.99					
7	Manikandan A	2/3	170.76					
6	Anirban	2/3	156.47					
8	K. Janani	1/3	94.17					
10	shreyas	1/3	91.44					

movie - 4K

110 GBs



8K

Now



8K → stored

4K → stored

2K → stored

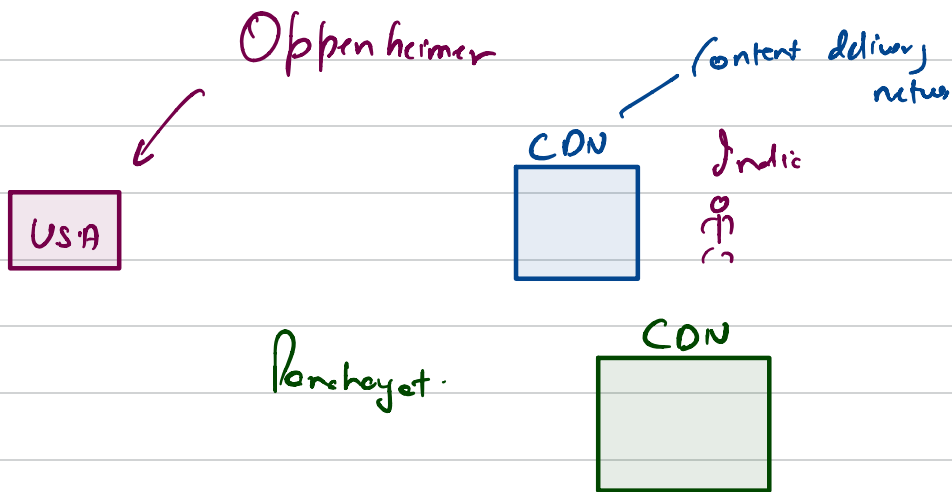
1K → stored

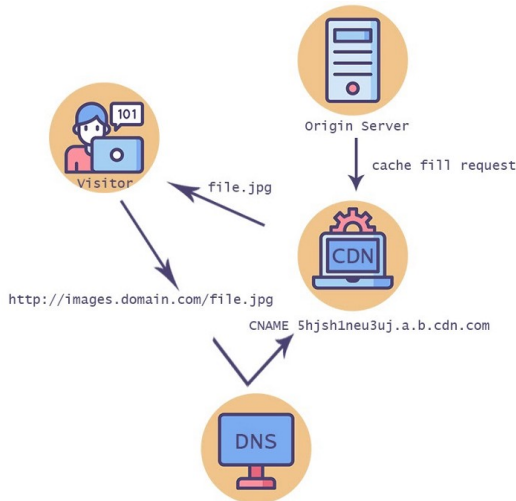
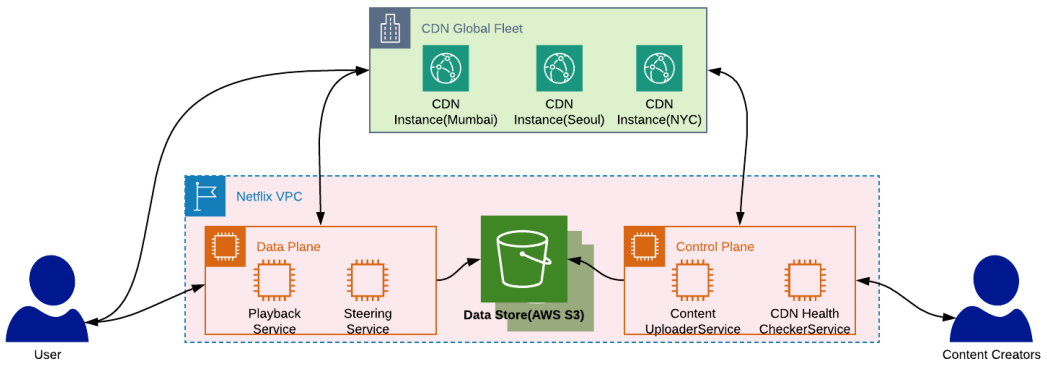
720p → stored

480

VIDEO

Video Resolution	Video Bitrate
240p	300 kbps
360p	500 kbps
HD 480p	1000 kbps
HD 720p	1500 kbps
HD 720p	2250 kbps
Full HD 1080p	3000 kbps
Full HD 1080p	4500 kbps
Quad HD 1440p	6000 kbps
Quad HD 1440p	9000 kbps
4K UHD 2160p	13,000 kbps
4K UHD 2160p	20,000 kbps



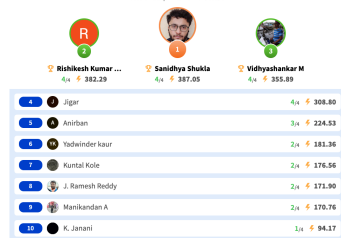


What is the role of a Content Delivery Network (CDN) in video streaming?

0 users have participated

- ☐ A To slow down the delivery of content to manage server load 0%
- ☐ B To provide a single centralized server for content storage 0%
- ☒ C To offer geographically distributed servers for fast content delivery 0%
- ☐ D To increase internet connection speeds universally 0%

End Quiz Now

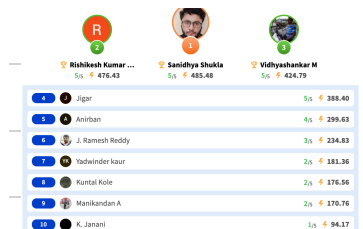


Which component of Netflix's recommendation system uses an item-item similarity matrix to suggest content?

0 users have participated

- ☐ A Trending Now algorithm 0%
- ☐ B Award-Winning Comedies selector 0%
- ☒ C Video-Video Similarity Ranker a.k.a Because You Watched (BYW) 0%
- ☐ D Continue Watching functionality 0%

End Quiz Now



	MOVIE-1	MOVIE-2	MOVIE-3	MOVIE-4
UID-1	LIKE -Yes TIMESTAMP 2:00:00 END No RECOMENDED Yes	LIKE -No TIMESTAMP 2:03:00 END Yes RECOMENDED No		
UID-2	LIKE Yes TIMESTAMP 1:58:21 END No RECOMENDED No	LIKE Yes TIMESTAMP 2:03:00 END Yes RECOMENDED Yes	LIKE No TIMESTAMP 1:20:00 END No RECOMENDED No	LIKE No TIMESTAMP 1:40:00 END Yes RECOMENDED Yes
UID-3	LIKE Yes TIMESTAMP 3:00:31 END Yes RECOMENDED Yes			

https://x.com/_trish_xD/status/1885662310091325737


Uber
 Mar 22, 2026, 12:00 AM

Data Scientist Onsite Machine Learning

Design an end-to-end machine learning approach to predict **driver acceptance probability** in a ride-sharing or delivery marketplace.

1. Why uber wants to predict this?
2. What DB might we need, what data might we need?
3. What model might you want to use?